Career Ka Dost

Ask before you decide

Class 10 Mathematics (Standard)

5 Sample Papers

Board Exam 2027 | CBSE 2026-27 syllabus | Latest 5-section paper pattern

5 full-length practice papers of 80 marks each (3 hours, 38 questions), put together from CBSE board questions of 2023 to 2026.

Paper	Difficulty
Paper 1	Easy
Paper 2	Moderate to Hard
Paper 3	Moderate to Hard
Paper 4	Moderate to Hard
Paper 5	Hard

Sample Paper 1

CLASS 10 | MATHEMATICS (STANDARD)

Board Examination Practice Paper (2026-27 pattern) | Level: Easy

Maximum Marks: 80

Time Allowed: 3 Hours

GENERAL INSTRUCTIONS

1. This question paper contains **38** questions. **All questions are compulsory.**
2. This question paper is divided into **five** sections - Section A, B, C, D and E.
3. **Section A:** Q. Nos. 1 to 18 are Multiple Choice Questions (MCQs) and Q. Nos. 19 and 20 are Assertion-Reason based questions of **1 mark** each.
4. **Section B:** Q. Nos. 21 to 25 are Very Short Answer (VSA) type questions carrying **2 marks** each.
5. **Section C:** Q. Nos. 26 to 31 are Short Answer (SA) type questions carrying **3 marks** each.
6. **Section D:** Q. Nos. 32 to 35 are Long Answer (LA) type questions carrying **5 marks** each.
7. **Section E:** Q. Nos. 36 to 38 are case study based questions carrying **4 marks** each. Internal choice is provided in one sub-part of the case study questions.
8. There is no overall choice. However, an internal choice has been provided in 2 questions in Section B (Q22, Q25), 2 in Section C (Q28, Q29), 2 in Section D (Q33, Q34) and 3 in Section E (Q36, Q37, Q38).
9. Draw neat diagrams wherever required. Take **$\pi = 22/7$** wherever required, if not stated.
10. Use of calculator is **NOT** allowed. Marks for each question are shown in square brackets on the right.

SECTION A - Multiple Choice and Assertion-Reason Questions (20 Marks)

1. If $(-1)^n + (-1)^8 = 0$, then n is : **[1]**

(A) any positive integer **(B)** any negative integer

(C) any odd number **(D)** any even number

2. For any natural number n, 5^n ends with the digit : **[1]**

(A) 0 **(B)** 5

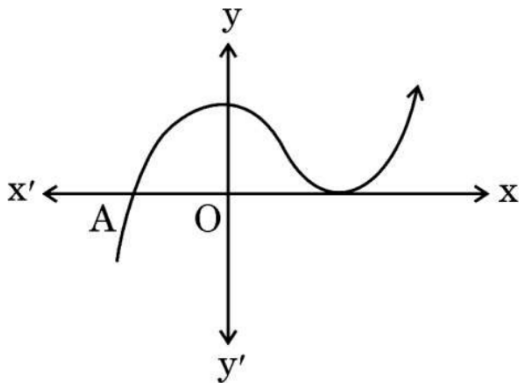
(C) 3 **(D)** 2

3. The LCM of 960 and 240 is : **[1]**

(A) 960 **(B)** 240

(C) 60 **(D)** 15

4. The graph of y = f(x) is given. The number of distinct zeroes of y = f(x) is : **[1]**



(A) 0 **(B)** 1

(C) 2 **(D)** 3

5. How many zeroes does $p(x) = (x - 2)(x + 3)$ have ? **[1]**

(A) Zero **(B)** One

(C) Two **(D)** Three

6. *n*th term of an A.P. is $7n + 4$. The common difference is : **[1]**

(A) $7n$ **(B)** 4

(C) 7 **(D)** 1

7. Three numbers in AP have the sum 30. What is its middle term ? **[1]**

(A) 4 **(B)** 10

(C) 16 **(D)** 8

8. The distance of the point A(4a, 3a) from x-axis is : **[1]**

(A) 3a **(B)** -3a

(C) 4a **(D)** -4a

9. If $\cos(\alpha + \beta) = 0$, then value of $\cos((\alpha + \beta)/2)$ is equal to : **[1]**

(A) $\frac{1}{\sqrt{2}}$ **(B)** $\frac{1}{2}$

(C) 0 **(D)** $\sqrt{2}$

10. For $\theta = 30°$, the value of $(2 \sin\theta \cos\theta)$ is : **[1]**

(A) 1 **(B)** $\frac{\sqrt{3}}{2}$

(C) $\frac{\sqrt{3}}{4}$ **(D)** $\frac{3}{2}$

11. If θ is an acute angle and $7 + 4\sin\theta = 9$, then the value of θ is : **[1]**

(A) 90° **(B)** 30°

(C) 45° **(D)** 60°

12. The tangents drawn at the extremities of the diameter of a circle are always : **[1]**

(A) parallel **(B)** perpendicular

(C) equal **(D)** intersecting

13. If the length of a chord of a circle is equal to its radius, then the angle subtended by chord at the centre is : **[1]**

(A) 60° **(B)** 30°

(C) 120° **(D)** 90°

14. The area of a semicircle of diameter 'd' is : **[1]**

(A) $\pi d^2/16$ **(B)** $\pi d^2/4$

(C) $\pi d^2/8$ **(D)** $\pi d^2/2$

15. The middle most observation of every data arranged in order is called : **[1]**

(A) mode **(B)** median

(C) mean **(D)** deviation

Career Ka Dost Ask before you decide

16. The mean and median of a frequency distribution are 43 and 43.4 respectively. The mode of the distribution is : **[1]**

(A) 43.4 **(B)** 42.4
(C) 44.2 **(D)** 49.3

17. If the probability of a player winning a game is 0.79, then the probability of his losing the same game is : **[1]**

(A) 1.79 **(B)** 0.31
(C) 0.21% **(D)** 0.21

18. Which of the following is not probability of an event ? **[1]**

(A) 0.89 **(B)** 52%
(C) $\frac{1}{13}$% **(D)** $\frac{1}{0.89}$

Assertion-Reason based questions. *Directions* : In Q. Nos. 19 and 20, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.

19. **Assertion (A) :** The tangents drawn at the end points of a diameter of a circle, are parallel. **[1]**
Reason (R) : Diameter of a circle is the longest chord.

20. **Assertion (A) :** The probability of selecting a number at random from the numbers 1 to 20 is 1. **[1]**
Reason (R) : For any event E, if P(E) = 1, then E is called a sure event.

SECTION B - Very Short Answer Questions (10 Marks)

21. Find the zeroes of the polynomial $p(x) = x^2 + \frac{4}{3}x - \frac{4}{3}$. **[2]**

22. (a) Solve the pair of equations x = 3 and y = -4 graphically. **[2]**

OR

(b) Using graphical method, find whether following system of linear equations is consistent or not :
x = 0 and y = -7

23. Solve the following system of linear equations : **[2]**
$2p + 3q = 13$ and $5p - 4q = -2$

24. If the distance between the points (4, p) and (1, 0) is 5, what is the value of p ? **[2]**

25. (a) If $x \cos 60° + y \cos 0° + \sin 30° - \cot 45° = 5$, then find the value of x + 2y. **[2]**

OR

(b) Evaluate : $\frac{\tan^2 60°}{\sin^2 60° + \cos^2 30°}$

SECTION C - Short Answer Questions (18 Marks)

26. Prove that $\sqrt{5}$ is an irrational number. **[3]**

27. Find the coordinates of the points of trisection of the line segment joining the points A(-1, 4) and B(-3, -2). **[3]**

28. (a) Prove that : $\frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta} = 1 + \sec\theta \operatorname{cosec}\theta$ **[3]**

OR

(b) Prove that : $\frac{\sin A + \cos A}{\sin A - \cos A} + \frac{\sin A - \cos A}{\sin A + \cos A} = \frac{2}{2\sin^2 A - 1}$

29. (a) Prove that the lengths of tangents drawn from an external point to a circle are equal. **[3]**

OR

(b) Two tangents TP and TQ are drawn to a circle with centre O from an external point T. Prove that $\angle PTQ = 2\angle OPQ$.

30. Two dice are thrown at the same time. Determine the probability that the (i) sum of the numbers on the two dice is 5, and (ii) difference of the numbers on the two dice is 3. **[3]**

31. Two different dice are thrown together. Find the probability that the numbers obtained have : **[3]**
(i) even sum,
(ii) even product.

SECTION D - Long Answer Questions (20 Marks)

32. Aarush bought 2 pencils and 3 chocolates for ₹ 11 and Tanish bought 1 pencil and 2 chocolates for ₹ 7 from the same shop. Represent this situation in the form of a pair of linear equations. Find the price of 1 pencil and 1 chocolate, graphically. **[5]**

33. (a) The perimeter of a right triangle is 60 cm and its hypotenuse is 25 cm. Find the lengths of other two sides of the triangle. **[5]**

OR

(b) A train travels a distance of 480 km at a uniform speed. If the speed had been 8 km/h less, then it would have taken 3 hours more to cover the same distance. Find the speed of the train.

34. (a) The diagonal BD of a parallelogram ABCD intersects the line segment AE at the point F, where E is any point on the side BC. Prove that DF × EF = FB × FA. **[5]**

OR

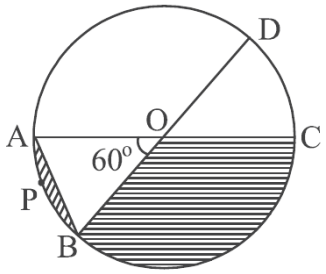
(b) In Δ ABC, if AD $\perp$ BC and AD^2 = BD × DC, then prove that $\angle$ BAC = 90°.

35. In the given figure, diameters AC and BD of the circle intersect at O. If ∠AOB = 60° and OA = 10 cm, then : **[5]**

(i) find the length of the chord AB.

(ii) find the area of shaded region.

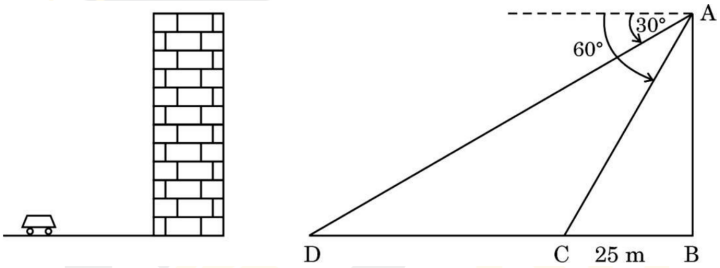
(Take π = 3.14 and √3 = 1.73)



SECTION E - Case Study Based Questions (12 Marks)

36. Read the case study given below and answer the questions that follow : **[4]**

Tejas is standing at the top of a building and observes a car at an angle of depression of 30° as it approaches the base of the building at a uniform speed. 6 seconds later, the angle of depression increases to 60°, and at that moment, the car is 25 m away from the building.



(36.1) What is the height of the building ? **[1]**

(36.2) What is the distance between the two positions of the car ? **[1]**

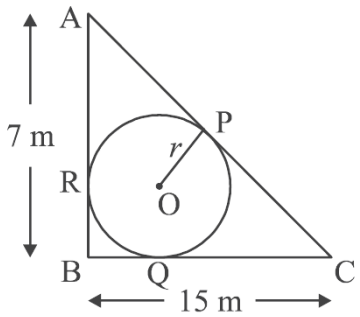
(36.3) (a) What would be the total time taken by the car to reach the foot of the building from the starting point ? **[2]**

OR

(36.3) (b) What is the distance of the observer from the car when it makes an angle of 60° ? **[2]**

37. Read the case study given below and answer the questions that follow : **[4]**

A backyard is in the shape of a triangle ABC with right angle at B. AB = 7 m and BC = 15 m. A circular pit was dug inside it such that it touches the walls AC, BC and AB at P, Q and R respectively such that AP = x m.



(37.1) Find the length of AR in terms of x. **[1]**

(37.2) Write the type of quadrilateral BQOR. **[1]**

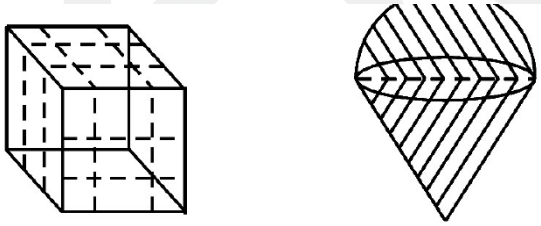
(37.3) (a) Find the length PC in terms of x and hence find the value of x. **[2]**

OR

(37.3) (b) Find x and hence find the radius r of circle. **[2]**

38. Read the case study given below and answer the questions that follow : **[4]**

On a Sunday your parents took you to a fair. You could see lot of toys displayed and you wanted them to buy a Rubik's cube and a strawberry ice-cream for you.



(38.1) Find the length of the diagonal of Rubik's cube if each edge measures 6 cm. **[1]**

(38.2) Find the volume of Rubik's cube if the length of the edge is 7 cm. **[1]**

(38.3) (a) What is the curved surface area of hemisphere (ice-cream) if the base radius is 7 cm ? **[2]**

OR

(38.3) (b) If two cubes of edges 4 cm are joined end-to-end, then find the surface area of the resulting cuboid. **[2]**

- End of Question Paper -

Sample Paper 2

CLASS 10 | MATHEMATICS (STANDARD)

Board Examination Practice Paper (2026-27 pattern) | Level: Moderate to Hard

Maximum Marks: 80

Time Allowed: 3 Hours

GENERAL INSTRUCTIONS

1. This question paper contains **38** questions. **All questions are compulsory.**
2. This question paper is divided into **five** sections - Section A, B, C, D and E.
3. **Section A:** Q. Nos. 1 to 18 are Multiple Choice Questions (MCQs) and Q. Nos. 19 and 20 are Assertion-Reason based questions of **1 mark** each.
4. **Section B:** Q. Nos. 21 to 25 are Very Short Answer (VSA) type questions carrying **2 marks** each.
5. **Section C:** Q. Nos. 26 to 31 are Short Answer (SA) type questions carrying **3 marks** each.
6. **Section D:** Q. Nos. 32 to 35 are Long Answer (LA) type questions carrying **5 marks** each.
7. **Section E:** Q. Nos. 36 to 38 are case study based questions carrying **4 marks** each. Internal choice is provided in one sub-part of the case study questions.
8. There is no overall choice. However, an internal choice has been provided in 2 questions in Section B (Q23, Q25), 2 in Section C (Q29, Q30), 2 in Section D (Q32, Q33) and 3 in Section E (Q36, Q37, Q38).
9. Draw neat diagrams wherever required. Take **$\pi = 22/7$** wherever required, if not stated.
10. Use of calculator is **NOT** allowed. Marks for each question are shown in square brackets on the right.

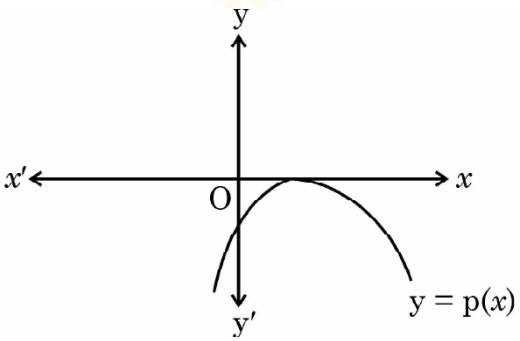
SECTION A - Multiple Choice and Assertion-Reason Questions (20 Marks)

1. The natural number 1 is : **[1]**

(A) a prime number. **(B)** a composite number.

(C) prime as well as composite. **(D)** neither prime nor composite.

2. The graph of $y = p(x)$ is given, for a polynomial $p(x)$. The number of zeroes of $p(x)$ from the graph is **[1]**



(A) 3 **(B)** 1

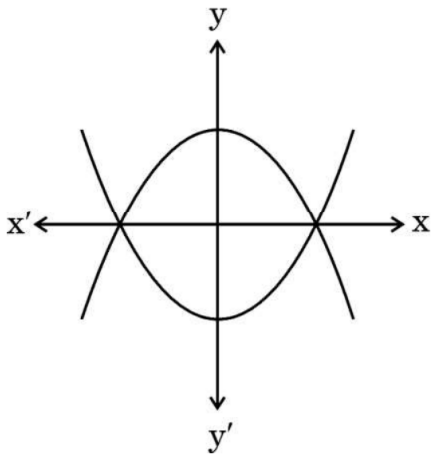
(C) 2 **(D)** 0

3. If the sum and the product of zeroes of a quadratic polynomial are $2\sqrt{3}$ and 3 respectively, then a quadratic polynomial is : **[1]**

(A) $x^2 + 2\sqrt{3}x - 3$ **(B)** $(x - \sqrt{3})^2$

(C) $x^2 - 2\sqrt{3}x - 3$ **(D)** $x^2 + 2\sqrt{3}x + 3$

4. Two polynomials are shown in the graph below. The number of distinct zeroes of both the polynomials is : **[1]**



(A) 3 **(B)** 5
(C) 2 **(D)** 4

5. If the roots of equation $ax^2 + bx + c = 0, a \neq 0$ are real and equal, then which of the following relation is true ? **[1]**

(A) $a = \frac{b^2}{c}$ **(B)** $b^2 = ac$

(C) $ac = \frac{b^2}{4}$ **(D)** $c = \frac{b^2}{a}$

6. If p - 1, p + 1 and 2p + 3 are in A.P., then the value of p is **[1]**

(A) -2 **(B)** 4
(C) 0 **(D)** 2

7. If a, b, c form an A.P. with common difference d, then the value of a - 2b - c is equal to **[1]**

(A) 2a + 4d **(B)** 0
(C) -2a - 4d **(D)** -2a - 3d

8. In an A.P., if the first term (a) = -16 and the common difference (d) = -2, then the sum of first 10 terms is : **[1]**

(A) - 200 **(B)** - 70
(C) - 250 **(D)** 250

9. Given cot θ = 3, the value of cos θ is : **[1]**

(A) $\frac{1}{3}$ **(B)** $1/\sqrt{10}$

(C) $3/\sqrt{10}$ **(D)** $\sqrt{10}/3$

10. A ladder 14 m long leans against a wall. If the foot of the ladder is 7 m from the wall, then the angle of elevation of the top of the wall is : **[1]**

(A) 15° **(B)** 30°
(C) 45° **(D)** 60°

11. The area of metal sheet required to make a closed hollow cylinder of height 2.4 m and base radius 0.7 m, is **[1]**

(A) 10.56 m^2 **(B)** 13.52 m^2

(C) 13.64 m^2 **(D)** 14.08 m^2

12. If the value of each observation of a statistical data is increased by 3, then the mean of the data **[1]**

(A) remains unchanged **(B)** increases by 3

(C) increases by 6 **(D)** increases by 3n

13. Mode and Mean of a data are 15x and 18x, respectively. Then the median of the data is : **[1]**

(A) x **(B)** 11x

(C) 17x **(D)** 34x

14. Probability of happening of an event is denoted by p and probability of non-happening of the event is denoted by q. Relation between p and q is **[1]**

(A) p + q = 1 **(B)** p = 1, q = 1

(C) p = q - 1 **(D)** p + q + 1 = 0

15. A girl calculates that the probability of her winning the first prize in a lottery is 0.08. If 6000 tickets are sold, how many tickets has she bought ? **[1]**

(A) 40 **(B)** 240

(C) 480 **(D)** 750

16. In a group of 20 people, 5 can't swim. If one person is selected at random, then the probability that he/she can swim, is **[1]**

(A) $\frac{3}{4}$ **(B)** $\frac{1}{3}$

(C) 1 **(D)** $\frac{1}{4}$

17. In a survey, it is found that every fifth person has a vehicle. The probability of a person NOT having a vehicle, is **[1]**

(A) $\frac{1}{5}$ **(B)** 5%

(C) $\frac{4}{5}$ **(D)** 95%

18. One card is drawn at random from a well shuffled deck of 52 playing cards. The probability that it is a red ace card, is : **[1]**

(A) $\frac{1}{13}$ **(B)** $\frac{1}{26}$

(C) $\frac{1}{52}$ **(D)** $\frac{1}{2}$

Career Ka Dost Ask before you decide

Assertion-Reason based questions. *Directions* : In Q. Nos. 19 and 20, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.

19. **Assertion (A) :** The mean of first 'n' natural numbers is $(n - 1)/2$. **[1]**
Reason (R) : The sum of first 'n' natural numbers is $n(n + 1)/2$.

20. **Assertion (A) :** If we join two hemispheres of same radius along their bases, then we get a sphere. **[1]**
Reason (R) : Total Surface Area of a sphere of radius r is $3\pi r^2$.

SECTION B - Very Short Answer Questions (10 Marks)

21. Find the greatest number which divides 85 and 72 leaving remainders 1 and 2 respectively. **[2]**

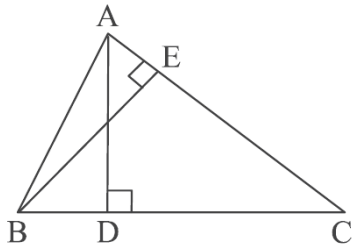
22. If the zeroes of the polynomial $x^2 + ax + b$ are in the ratio 3 : 4, then prove that $12a^2 = 49b$. **[2]**

23. (a) Solve the pair of equations $x = 5$ and $y = 7$ graphically. **[2]**

OR

(b) Using graphical method, find whether pair of equations $x = 0$ and $y = -3$, is consistent or not.

24. In ΔABC, altitudes AD and BE are drawn. If AD = 7 cm, BE = 9 cm and EC = 12 cm then, find the length of CD. **[2]**



25. (a) If $\tan A = \frac{4}{3}$, find sin A and cos A. **[2]**

OR

(b) Express cos A and tan A in terms of sin A.

SECTION C - Short Answer Questions (18 Marks)

26. In a teachers' workshop, the number of teachers teaching French, Hindi and English are 48, 80 and 144 respectively. Find the minimum number of rooms required if in each room the same number of teachers are seated and all of them are of the same subject. **[3]**

27. Find the ratio in which the y-axis divides the line segment joining the points (5, -6) and (-1, -4). Also find the point of intersection. **[3]**

28. The three vertices of a rhombus PQRS are P(2, -3), Q(6, 5) and R(-2, 1). Find the coordinates of the fourth vertex S and coordinates of the point where both the diagonals PR and QS intersect. **[3]**

29. (a) Prove that : **[3]**

$$\frac{\sec^3\theta}{\sec^2\theta - 1} + \frac{\text{cosec}^3\theta}{\text{cosec}^2\theta - 1} = \sec\theta \cdot \text{cosec}\,\theta\,(\sec\theta + \text{cosec}\,\theta)$$

OR

(b) If $\frac{\sec\alpha}{\text{cosec}\,\beta} = p$ and $\frac{\tan\alpha}{\text{cosec}\,\beta} = q$, then prove that $(p^2 - q^2)\sec^2\alpha = p^2$.

30. (a) Prove that the tangent at any point of a circle is perpendicular to the radius through the point of contact. **[3]**

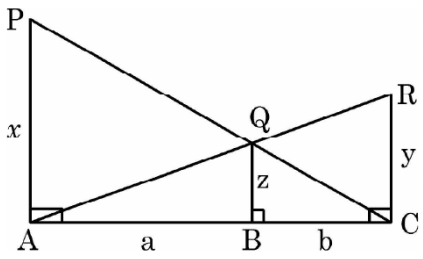
OR

(b) If a regular hexagon ABCDEF circumscribes a circle, then prove that AB + CD + EF = BC + DE + FA.

31. Find the area of the sector of a circle of radius 42 cm and of central angle 30°. Also, find the area of the corresponding major sector. [Use $\pi = \frac{22}{7}$] **[3]**

SECTION D - Long Answer Questions (20 Marks)

32. (a) PA, QB and RC are each perpendicular to AC. If AP = *x*, QB = z, RC = y, AB = a and BC = b, then prove that $\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$. **[5]**

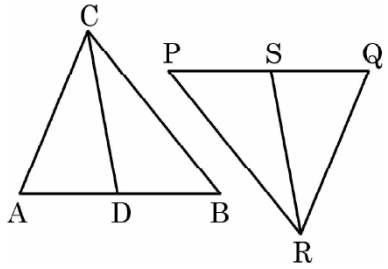


OR

(b) In the given figure, CD and RS are respectively the medians of ΔABC and ΔPQR. If ΔABC ~ ΔPQR then prove that :

(i) ΔADC ~ ΔPSR

(ii) AD × PR = AC × PS

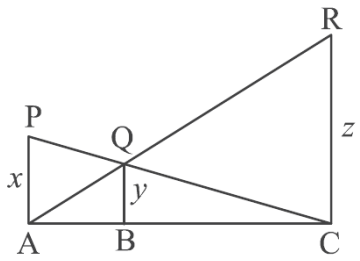


33. (a) If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then prove that the other two sides are divided in the same ratio. **[5]**

OR

(b) In the given figure PA, QB and RC are each perpendicular to AC. If AP = x, BQ = y and CR = z, then prove that

$$\frac{1}{x} + \frac{1}{z} = \frac{1}{y}$$



34. From the top of a 15 m high building, the angle of elevation of the top of a tower is found to be 30°. From the bottom of the same building, the angle of elevation of the top of the tower is found to be 60°. Find the height of the tower and the distance between tower and the building. **[5]**

35. A vessel is in the form of a hollow hemisphere mounted by a hollow cylinder. The diameter of the hemisphere is 14 cm and the total height of the vessel is 13 cm. Find the inner surface area and the volume of the vessel. **[5]**

SECTION E - Case Study Based Questions (12 Marks)

36. Read the case study given below and answer the questions that follow : **[4]**

> A rectangular floor area can be completely tiled with 200 square tiles. If the side length of each tile is increased by 1 unit, it would take only 128 tiles to cover the floor.

(36.1) Assuming the original length of each side of a tile be x units, make a quadratic equation from the above information. **[1]**

(36.2) Write the corresponding quadratic equation in standard form. **[1]**

(36.3) (a) Find the value of x, the length of side of a tile by factorisation. **[2]**

OR

(36.3) (b) Solve the quadratic equation for x, using quadratic formula. **[2]**

37. Read the case study given below and answer the questions that follow : **[4]**

> Your elder brother wants to buy a car and plans to take a loan from a bank for his car. He repays his total loan of ₹ 1,18,000 by paying every month, starting with the first instalment of ₹ 1,000 and he increases the instalment by ₹ 100 every month.

(37.1) Find the amount paid by him in the 30th instalment. **[1]**

(37.2) If the total number of instalments is 40, what is the amount paid in the last instalment ? **[1]**

(37.3) (a) What amount does he still have to pay after the 30th instalment ? **[2]**

OR

(37.3) (b) Find the ratio of the tenth instalment to the last instalment. **[2]**

38. Read the case study given below and answer the questions that follow : **[4]**

BINGO is game of chance. The host has 75 balls numbered 1 through 75. Each player has a BINGO card with some numbers written on it. The participant cancels the number on the card when called out a number written on the ball selected at random. Whosoever cancels all the numbers on his/her card, says BINGO and wins the game.
The table given below, shows the data of one such game where 48 balls were used before Tara said 'BINGO'.

Numbers announced	Number of times
0-15	8
15-30	9
30-45	10
45-60	12
60-75	9

(38.1) Write the median class. **[1]**

(38.2) When first ball was picked up, what was the probability of calling out an even number ? **[1]**

(38.3) (a) Find median of the given data. **[2]**

OR

(38.3) (b) Find mode of the given data. **[2]**

- End of Question Paper -

Career Ka Dost
Ask before you decide

Sample Paper 3

CLASS 10 | MATHEMATICS (STANDARD)

Board Examination Practice Paper (2026-27 pattern) | Level: Moderate to Hard

Maximum Marks: 80

Time Allowed: 3 Hours

GENERAL INSTRUCTIONS

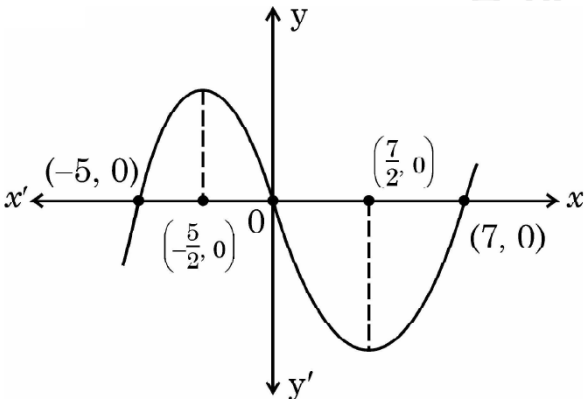
1. This question paper contains **38** questions. **All questions are compulsory.**
2. This question paper is divided into **five** sections - Section A, B, C, D and E.
3. **Section A:** Q. Nos. 1 to 18 are Multiple Choice Questions (MCQs) and Q. Nos. 19 and 20 are Assertion-Reason based questions of **1 mark** each.
4. **Section B:** Q. Nos. 21 to 25 are Very Short Answer (VSA) type questions carrying **2 marks** each.
5. **Section C:** Q. Nos. 26 to 31 are Short Answer (SA) type questions carrying **3 marks** each.
6. **Section D:** Q. Nos. 32 to 35 are Long Answer (LA) type questions carrying **5 marks** each.
7. **Section E:** Q. Nos. 36 to 38 are case study based questions carrying **4 marks** each. Internal choice is provided in one sub-part of the case study questions.
8. There is no overall choice. However, an internal choice has been provided in 2 questions in Section B (Q24, Q25), 2 in Section C (Q27, Q29), 2 in Section D (Q32, Q33) and 3 in Section E (Q36, Q37, Q38).
9. Draw neat diagrams wherever required. Take **π = 22/7** wherever required, if not stated.
10. Use of calculator is **NOT** allowed. Marks for each question are shown in square brackets on the right.

SECTION A - Multiple Choice and Assertion-Reason Questions (20 Marks)

1. Which of the following is a quadratic polynomial with zeroes $\frac{5}{3}$ and 0 ? **[1]**

(A) $3x(3x - 5)$
(B) $3x(x - 5)$
(C) $x^2 - \frac{5}{3}$
(D) $\frac{5}{3}x^2$

2. The graph of y = p(x) is given in the adjoining figure. Zeroes of the polynomial p(x) are **[1]**



(A) -5, 7
(B) $\frac{-5}{2}, \frac{-7}{2}$
(C) -5, 0, 7
(D) $-5, \frac{-5}{2}, \frac{7}{2}, 7$

3. If the sum of zeroes of the polynomial $p(x) = 2x^2 - k\sqrt{2}\,x + 1$ is $\sqrt{2}$, then value of k is : **[1]**

(A) $\sqrt{2}$ **(B)** 2

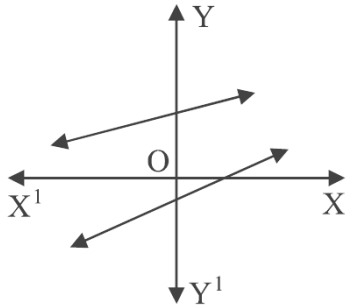
(C) $2\sqrt{2}$ **(D)** $\frac{1}{2}$

4. The value of k for which the pair of equations kx = y + 2 and 6x = 2y + 3 has infinitely many solutions, **[1]**

(A) is k = 3 **(B)** does not exist

(C) is k = -3 **(D)** is k = 4

5. In the given figure, graphs of two linear equations are shown. The pair of these linear equations is : **[1]**



(A) consistent with unique solution.

(B) consistent with infinitely many solutions.

(C) inconsistent.

(D) inconsistent but can be made consistent by extending these lines.

6. If the pair of linear equations : $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ is consistent and dependent, then **[1]**

(A) $a_1/a_2 \neq b_1/b_2$ **(B)** $a_1/a_2 \neq b_1/b_2 = c_1/c_2$

(C) $a_1/a_2 = b_1/b_2 \neq c_1/c_2$ **(D)** $a_1/a_2 = b_1/b_2 = c_1/c_2$

7. In triangles ABC and DEF, ∠ B = ∠ E, ∠ F = ∠ C and AB = 3 DE. Then, the two triangles are : **[1]**

(A) congruent but not similar **(B)** congruent as well as similar

(C) neither congruent nor similar **(D)** similar but not congruent

8. The distance between the points $(0, 2\sqrt{5})$ and $(-2\sqrt{5}, 0)$ is **[1]**

(A) $2\sqrt{10}$ units **(B)** $4\sqrt{10}$ units

(C) $2\sqrt{20}$ units **(D)** 0

9. The centre of a circle is at (2, 0). If one end of a diameter is at (6, 0), then the other end is at : **[1]**

(A) (0, 0) **(B)** (4, 0)

(C) (-2, 0) **(D)** (-6, 0)

10. The mid-point of the line segment joining the points P(-4, 5) and Q(4, 6) lies on : **[1]**

(A) x-axis **(B)** y-axis

(C) origin **(D)** neither x-axis nor y-axis

11. If 2 sin A = 1, then the value of tan A + cot A is : **[1]**

(A) $\sqrt{3}$ **(B)** $4/\sqrt{3}$

(C) $\sqrt{3}/2$ **(D)** 1

12. A kite is flying at a height of 150 m from the ground. It is attached to a string inclined at an angle of 30° to the horizontal. The length of the string is : **[1]**

(A) 100√3 m **(B)** 300 m

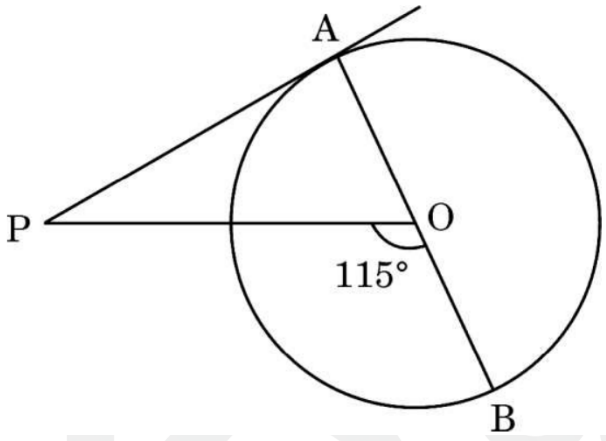
(C) 150√2 m **(D)** 150√3 m

13. From a point on the ground, which is 60 m away from the foot of a vertical tower, the angle of elevation of the top of the tower is found to be 45°. The height (in metres) of the tower is : **[1]**

(A) 10√3 **(B)** 30√3

(C) 60 **(D)** 30

14. In the given figure, PA is a tangent from an external point P to a circle with centre O. If ∠ POB = 115°, then ∠ APO is equal to : **[1]**



(A) 25° **(B)** 65°

(C) 90° **(D)** 35°

15. An arc of a circle is of length 5π cm and the sector it bounds has an area of 20π cm^2. Its radius is : **[1]**

(A) 10 cm **(B)** 1 cm

(C) 5 cm **(D)** 8 cm

16. Curved surface area of a cylinder of height 5 cm is 94.2 cm^2. Radius of the cylinder is (Take π = 3.14) **[1]**

(A) 2 cm **(B)** 3 cm

(C) 2.9 cm **(D)** 6 cm

17. A solid sphere is cut into two hemispheres. The ratio of the surface areas of sphere to that of two hemispheres taken together, is : **[1]**

(A) 1 : 1 **(B)** 1 : 4

(C) 2 : 3 **(D)** 3 : 2

18. The probability for a randomly selected number out of 1, 2, 3, 4, ..., 25 to be a composite number is : **[1]**

(A) $\frac{15}{25}$ **(B)** $\frac{10}{25}$

(C) $\frac{11}{25}$ **(D)** $\frac{9}{25}$

Assertion-Reason based questions. *Directions* : In Q. Nos. 19 and 20, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

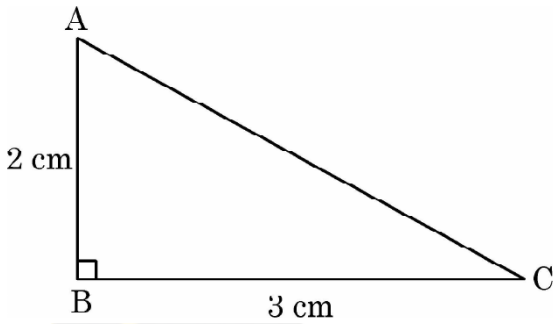
(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

19. **Assertion (A) :** The perimeter of ΔABC is a rational number. **[1]**

Reason (R) : The sum of the squares of two rational numbers is always rational.



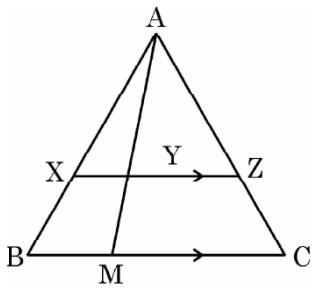
20. **Assertion (A) :** The surface area of the cuboid formed by joining two cubes of sides 4 cm each, end-to-end, is 160 cm^2. **[1]**

Reason (R) : The surface area of a cuboid of dimensions $l \times b \times h$ is $(lb + bh + hl)$.

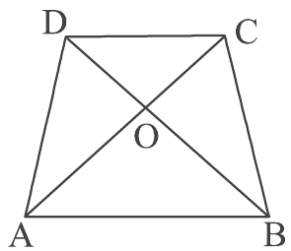
SECTION B - Very Short Answer Questions **(10 Marks)**

21. Find the least number which when divided by 12, 16 and 24 leaves remainder 7 in each case. **[2]**

22. In the given figure, XZ is parallel to BC. AZ = 3 cm, ZC = 2 cm, BM = 3 cm and MC = 5 cm. Find the length of XY. **[2]**



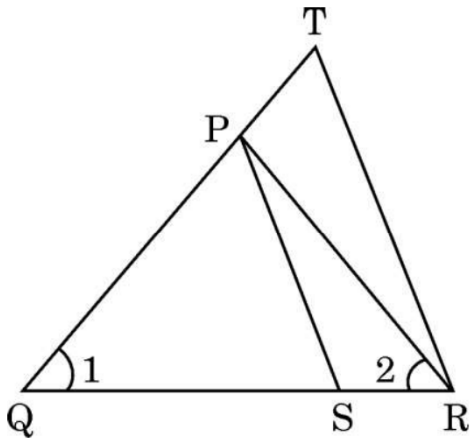
23. Diagonals AC and BD of a trapezium ABCD intersect at O, where AB||DC. If $\frac{DO}{OB} = \frac{1}{2}$, then show that AB = 2CD **[2]**



24. (a) If Δ ABC ~ Δ PQR in which AB = 6 cm, BC = 4 cm, AC = 8 cm and PR = 6 cm, then find the length of (PQ + QR). **[2]**

OR

(b) In the given figure, $\frac{QR}{QS} = \frac{QT}{PR}$ and ∠ 1 = ∠ 2, show that Δ PQS ~ Δ TQR.



25. (a) Evaluate : **[2]**

$$\frac{5\cos^2 60° + 4\sec^2 30° - \tan^2 45°}{\sin^2 30° + \cos^2 30°}$$

OR

(b) Prove that :

$$1 + \frac{\cot^2 \alpha}{1 + \text{cosec } \alpha} = \text{cosec } \alpha$$

SECTION C - Short Answer Questions (18 Marks)

26. Three sets of Physics, Chemistry and Mathematics books have to be stacked in such a way that all the books are stored subject-wise and the height of each stack is the same. The number of Physics books is 144, the number of Chemistry books is 180 and the number of Mathematics books is 192. Assuming that the books are of same thickness, determine the number of stacks of Physics, Chemistry and Mathematics books. **[3]**

27. (a) Find the ratio in which the point $(\frac{8}{5}, y)$ divides the line segment joining the points (1, 2) and (2, 3). Also, find the value of y. **[3]**

OR

(b) ABCD is a rectangle formed by the points A (-1, -1), B (-1, 6), C (3, 6) and D (3, -1). P, Q, R and S are mid-points of sides AB, BC, CD and DA respectively. Show that diagonals of the quadrilateral PQRS bisect each other.

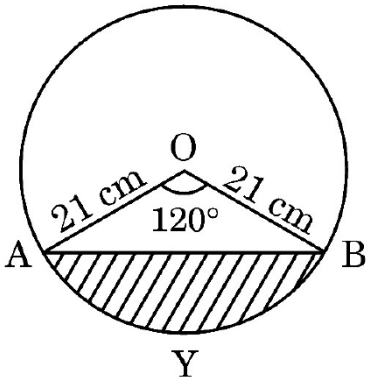
28. Prove that : $2(\sin^6 \theta + \cos^6 \theta) - 3(\sin^4 \theta + \cos^4 \theta) + 1 = 0$. **[3]**

29. (a) Prove that the lengths of tangents drawn from an external point to a circle are equal. **[3]**

OR

(b) Two tangents TP and TQ are drawn to a circle with centre O from an external point T. Prove that ∠PTQ = 2∠OPQ.

30. Find the area of the segment AYB shown in the figure, if the radius of the circle is 21 cm and ∠AOB = 120°. [Use $\pi = \frac{22}{7}$] **[3]**



31. A wooden toy is made by scooping out a hemisphere of same radius as of cylinder, from each end of a wooden solid cylinder. If the height of the cylinder is 20 cm and its base is of radius 7 cm, find the total surface area of the toy. **[3]**

SECTION D - Long Answer Questions (20 Marks)

32. (a) A two-digit number is such that the product of its digits is 12. When 36 is added to this number, the digits interchange their places. Find the number. **[5]**

OR

(b) A student scored a total of 32 marks in class tests in Mathematics and Science. Had he scored 2 marks less in Science and 4 marks more in Mathematics, the product of his marks would have been 253. Find his marks in the two subjects.

33. (a) The sum of first and eighth terms of an A.P. is 32 and their product is 60. Find the first term and common difference of the A.P. Hence, also find the sum of its first 20 terms. **[5]**

OR

(b) In an A.P. of 40 terms, the sum of first 9 terms is 153 and the sum of last 6 terms is 687. Determine the first term and common difference of A.P. Also, find the sum of all the terms of the A.P.

34. Find the missing frequency 'f' in the following table, if the mean of the given data is 18. Hence find the mode. **[5]**

Daily Allowance	Number of Children
11 – 13	7
13 – 15	6
15 – 17	9
17 – 19	13
19 – 21	f
21 – 23	5
23 – 25	4

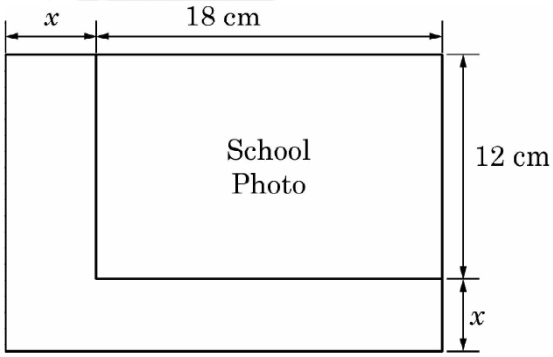
35. If the median of the following distribution is 32.5, then find the values of x and y. **[5]**

Class	Frequency
0 – 10	x
10 – 20	5
20 – 30	9
30 – 40	12
40 – 50	y
50 – 60	3
60 – 70	2
Total	40

SECTION E - Case Study Based Questions (12 Marks)

36. Read the case study given below and answer the questions that follow : **[4]**

While designing the school year book, a teacher asked the student that the length and width of a particular photo is increased by x units each to double the area of the photo. The original photo is 18 cm long and 12 cm wide.
Based on the above information, answer the following questions :



(36.1) Write an algebraic equation depicting the above information. **[1]**

(36.2) Write the corresponding quadratic equation in standard form. **[1]**

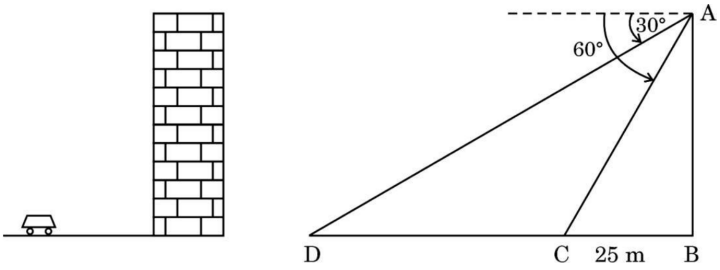
(36.3) (a) What should be the new dimensions of the enlarged photo ? **[2]**

OR

(36.3) (b) Can any rational value of x make the new area equal to 220 cm^2 ? **[2]**

37. Read the case study given below and answer the questions that follow : **[4]**

Tejas is standing at the top of a building and observes a car at an angle of depression of 30° as it approaches the base of the building at a uniform speed. 6 seconds later, the angle of depression increases to 60°, and at that moment, the car is 25 m away from the building.



(37.1) What is the height of the building ? **[1]**

(37.2) What is the distance between the two positions of the car ? **[1]**

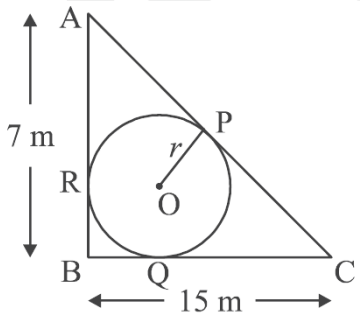
(37.3) (a) What would be the total time taken by the car to reach the foot of the building from the starting point ? **[2]**

OR

(37.3) (b) What is the distance of the observer from the car when it makes an angle of 60° ? **[2]**

38. Read the case study given below and answer the questions that follow : **[4]**

A backyard is in the shape of a triangle ABC with right angle at B. AB = 7 m and BC = 15 m. A circular pit was dug inside it such that it touches the walls AC, BC and AB at P, Q and R respectively such that AP = x m.



(38.1) Find the length of AR in terms of x. **[1]**

(38.2) Write the type of quadrilateral BQOR. **[1]**

(38.3) (a) Find the length PC in terms of x and hence find the value of x. **[2]**

OR

(38.3) (b) Find x and hence find the radius r of circle. **[2]**

- End of Question Paper -

Sample Paper 4

CLASS 10 | MATHEMATICS (STANDARD)

Board Examination Practice Paper (2026-27 pattern) | Level: Moderate to Hard

Maximum Marks: 80

Time Allowed: 3 Hours

GENERAL INSTRUCTIONS

1. This question paper contains **38** questions. **All questions are compulsory.**
2. This question paper is divided into **five** sections - Section A, B, C, D and E.
3. **Section A:** Q. Nos. 1 to 18 are Multiple Choice Questions (MCQs) and Q. Nos. 19 and 20 are Assertion-Reason based questions of **1 mark** each.
4. **Section B:** Q. Nos. 21 to 25 are Very Short Answer (VSA) type questions carrying **2 marks** each.
5. **Section C:** Q. Nos. 26 to 31 are Short Answer (SA) type questions carrying **3 marks** each.
6. **Section D:** Q. Nos. 32 to 35 are Long Answer (LA) type questions carrying **5 marks** each.
7. **Section E:** Q. Nos. 36 to 38 are case study based questions carrying **4 marks** each. Internal choice is provided in one sub-part of the case study questions.
8. There is no overall choice. However, an internal choice has been provided in 2 questions in Section B (Q23, Q24), 2 in Section C (Q29, Q30), 2 in Section D (Q32, Q33) and 3 in Section E (Q36, Q37, Q38).
9. Draw neat diagrams wherever required. Take **$\pi = 22/7$** wherever required, if not stated.
10. Use of calculator is **NOT** allowed. Marks for each question are shown in square brackets on the right.

SECTION A - Multiple Choice and Assertion-Reason Questions (20 Marks)

1. If two positive integers p and q can be expressed as $p = 18\,a^2b^4$ and $q = 20\,a^3b^2$, where a and b are prime numbers, then LCM (p, q) is : **[1]**

(A) $2\,a^2b^2$ **(B)** $180\,a^2b^2$

(C) $12\,a^2b^2$ **(D)** $180\,a^3b^4$

2. If HCF(98, 28) = m and LCM(98, 28) = n, then the value of n - 7m is : **[1]**

(A) 0 **(B)** 28

(C) 98 **(D)** 198

3. Which of the following is a rational number between $\sqrt{3}$ and $\sqrt{5}$? **[1]**

(A) 1.4142387954012 .... **(B)** $2.32\bar{6}$

(C) π **(D)** 1.857142

4. If α, β are zeroes of a polynomial $p(x) = 2x^2 - x - 1$ then $\alpha^2 + \beta^2$ is equal to **[1]**

(A) $\frac{-3}{4}$ **(B)** $\frac{5}{4}$

(C) $\frac{1}{4}$ **(D)** $\frac{3}{4}$

5. The zeroes of a polynomial $x^2 + px + q$ are twice the zeroes of the polynomial $4x^2 - 5x - 6$. The value of p is : **[1]**

(A) $-\frac{5}{2}$ **(B)** $\frac{5}{2}$

(C) - 5 **(D)** 10

6. If the sum of first m terms of an AP is $2m^2 + 3m$, then its second term is : **[1]**

(A) 10 **(B)** 9

(C) 12 **(D)** 4

7. The value of x for which 2x, (x + 10) and (3x + 2) are the three consecutive terms of an A.P. is : **[1]**

(A) 6 **(B)** - 6

(C) 18 **(D)** - 18

8. In triangles ABC and PQR, $\angle A = \angle Q$ and $\angle B = \angle R$, then AB : AC is equal to : **[1]**

(A) PQ : PR **(B)** PQ : QR

(C) QR : QP **(D)** PR : QR

9. The end-points of a diameter of a circle are (2, 4) and (-3, -1). The radius of the circle is **[1]**

(A) $2\sqrt{5}$ **(B)** $\frac{5}{2}\sqrt{5}$

(C) $\frac{5}{2}\sqrt{2}$ **(D)** $5\sqrt{2}$

10. If the distance between the points (4, p) and (1, 0) is 5, then p is equal to : **[1]**

(A) ± 4 **(B)** 4

(C) - 4 **(D)** 0

11. $[\frac{3}{4}\tan^2 30° - \sec^2 45° + \sin^2 60°]$ is equal to **[1]**

(A) -1 **(B)** $\frac{5}{6}$

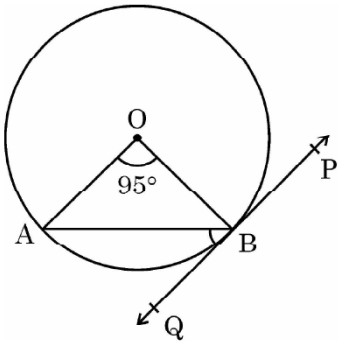
(C) $\frac{-3}{2}$ **(D)** $\frac{1}{6}$

12. The value of $\tan^2 \theta - (\frac{1}{\cos \theta} \times \sec \theta)$ is : **[1]**

(A) 1 **(B)** 0

(C) - 1 **(D)** 2

13. In the given figure, PQ is tangent to the circle centred at O. If $\angle AOB = 95°$, then the measure of $\angle ABQ$ will be **[1]**



(A) 47.5° **(B)** 42.5°

(C) 85° **(D)** 95°

14. If a sector of a circle has an area of 40π sq. units and a central angle of 72°, the radius of the circle is : **[1]**

(A) 200 units **(B)** 100 units

(C) 20 units **(D)** $10\sqrt{2}$ units

15. The curved surface area of a cone having height 24 cm and radius 7 cm, is **[1]**

(A) 528 cm^2 **(B)** 1056 cm^2

(C) 550 cm^2 **(D)** 500 cm^2

16. The volume of the largest right circular cone that can be carved out from a solid cube of edge 2 cm is : **[1]**

(A) $\frac{4\pi}{3}$ cu cm **(B)** $\frac{5\pi}{3}$ cu cm

(C) $\frac{8\pi}{3}$ cu cm **(D)** $\frac{2\pi}{3}$ cu cm

17. For some data $x_1, x_2, \ldots\ldots x_n$ with respective frequencies $f_1, f_2, \ldots\ldots f_n$, the value of $\Sigma f_i (x_i - \bar{x})$ (sum from 1 to n) is equal to : **[1]**

(A) $n\bar{x}$ **(B)** 1

(C) Σf_i **(D)** 0

18. Two dice are rolled together. The probability of getting sum of numbers on the two dice as 2, 3 or 5, is : **[1]**

(A) $\frac{7}{36}$ **(B)** $\frac{11}{36}$

(C) $\frac{5}{36}$ **(D)** $\frac{4}{9}$

Assertion-Reason based questions. *Directions* : In Q. Nos. 19 and 20, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

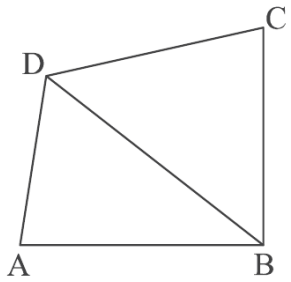
19. **Assertion (A) :** Point P(0, 2) is the point of intersection of y-axis with the line 3x + 2y = 4. **[1]**
Reason (R) : The distance of point P(0, 2) from x-axis is 2 units.

20. **Assertion (A) :** The tangents drawn at the end points of a diameter of a circle, are parallel. **[1]**
Reason (R) : Diameter of a circle is the longest chord.

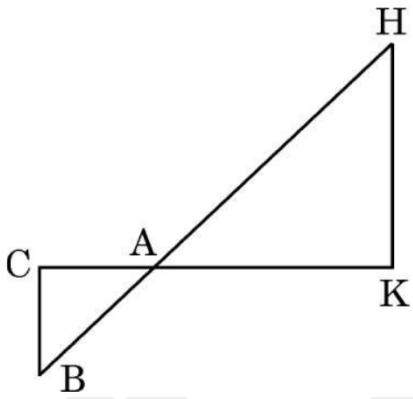
SECTION B - Very Short Answer Questions (10 Marks)

21. Solve the following system of linear equations algebraically : **[2]**
2x + 5y = -4; 4x - 3y = 5

22. In the given figure, ABCD is a quadrilateral. Diagonal BD bisects ∠B and ∠D both. **[2]**
Prove that :
(i) ΔABD ~ ΔCBD
(ii) AB = BC

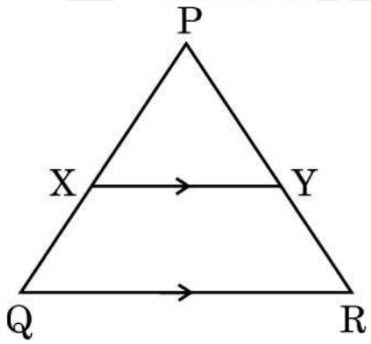


23. (a) In the given figure, ΔAHK ~ ΔABC. If AK = 10 cm, BC = 3.5 cm and HK = 7 cm, find the length of AC. **[2]**



OR

(b) In the given figure, XY ∥ QR, PQ/XQ = $\frac{7}{3}$ and PR = 6.3 cm. Find the length of YR.

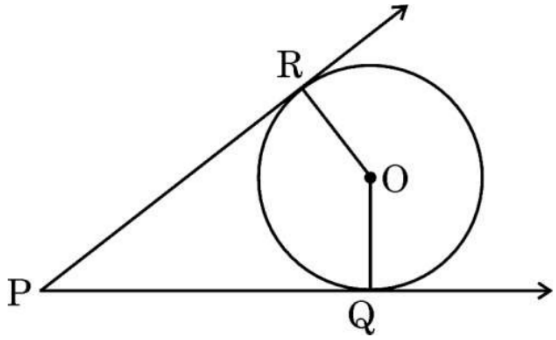


24. (a) If sin θ + cos θ = √3 , then find the value of sin θ · cos θ. **[2]**

OR

(b) If sin α = 1/√2 and cot β = √3 , then find the value of cosec α + cosec β.

25. In the given figure, O is the centre of the circle. PQ and PR are tangents. Show that the quadrilateral PQOR is cyclic. **[2]**



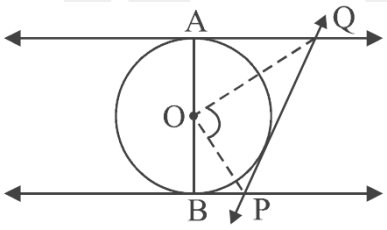
SECTION C - Short Answer Questions (18 Marks)

26. In a teachers' workshop, the number of teachers teaching French, Hindi and English are 48, 80 and 144 respectively. Find the minimum number of rooms required if in each room the same number of teachers are seated and all of them are of the same subject. **[3]**

27. If (-5, 3) and (5, 3) are two vertices of an equilateral triangle, then find co-ordinates of the third vertex, given that origin lies inside the triangle. (Take $\sqrt{3} = 1.7$) **[3]**

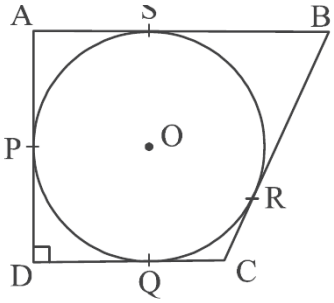
28. Prove that : $\frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta} = 1 + \sec\theta \operatorname{cosec}\theta$ **[3]**

29. (a) In the given figure, AB is a diameter of the circle with centre O. AQ, BP and PQ are tangents to the circle. Prove that $\angle POQ = 90°$. **[3]**

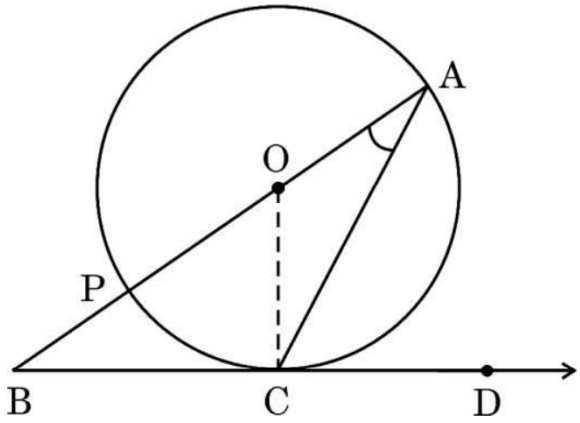


OR

(b) A circle with centre O and radius 8 cm is inscribed in a quadrilateral ABCD in which P, Q, R, S are the points of contact as shown. If AD is perpendicular to DC, BC = 30 cm and BS = 24 cm, then find the length DC.



30. (a) In the given figure, O is the centre of the circle and BCD is tangent to it at C. Prove that ∠ BAC + ∠ ACD = 90°. **[3]**



OR

(b) Prove that opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre of the circle.

31. A room is in the form of a cylinder surmounted by a hemispherical dome. The base radius of the hemisphere is half of the height of the cylindrical part. If the room contains $\frac{1408}{21}$ m^3 of air, find the height of the cylindrical part. (Use $\pi = \frac{22}{7}$). **[3]**

SECTION D - Long Answer Questions **(20 Marks)**

32. (a) In a class test, the sum of Anamika's marks obtained in Maths and Science is 30. Had she got 2 marks more in Maths and 3 marks less in Science, the product of the marks would have been 210. Find the marks she got in the two subjects. **[5]**

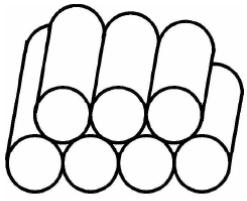
OR

(b) The length of hypotenuse (in cm) of a right-angled triangle is 6 cm more than twice the length of its shortest side. If the length of its third side is 6 cm less than thrice the length of its shortest side, find the dimensions of the triangle.

33. (a) The ratio of the 11th term to 17th term of an A.P. is 3 : 4. Find the ratio of 5th term to 21st term of the same A.P. Also, find the ratio of the sum of first 5 terms to that of first 21 terms. **[5]**

OR

(b) 250 logs are stacked in the following manner :
22 logs in the bottom row, 21 in the next row, 20 in the row next to it and so on (as shown by an example). In how many rows, are the 250 logs placed and how many logs are there in the top row ?



(Example)

34. An aeroplane when flying at a height of 3000 m from the ground passes vertically above another aeroplane at an instant when the angles of elevation of the two planes from the same point on the ground are 60° and 45° respectively. Find the vertical distance between the aeroplanes at that instant. Also, find the distance of first plane from the point of observation. (Take $\sqrt{3} = 1.73$) **[5]**

35. The following frequency distribution gives the monthly consumption of electricity of 68 consumers of a locality. Find the mean and mode of the data : **[5]**

Monthly Consumption (in units)	Number of Consumers
65 – 85	4
85 – 105	5
105 – 125	13
125 – 145	20
145 – 165	14
165 – 185	8
185 – 205	4

SECTION E - Case Study Based Questions **(12 Marks)**

36. Read the case study given below and answer the questions that follow : **[4]**

A school is organizing a charity run to raise funds for a local hospital. The run is planned as a series of rounds around a track, with each round being 300 metres. To make the event more challenging and engaging, the organizers decide to increase the distance of each subsequent round by 50 metres. For example, the second round will be 350 metres, the third round will be 400 metres and so on. The total number of rounds planned is 10.

(36.1) Write the fourth, fifth and sixth term of the Arithmetic Progression so formed. **[1]**

(36.2) Determine the distance of the 8th round. **[1]**

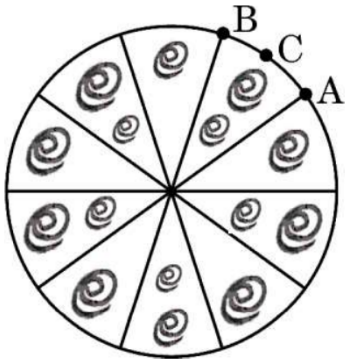
(36.3) (a) Find the total distance run after completing all 10 rounds. **[2]**

OR

(36.3) (b) If a runner completes only the first 6 rounds, what is the total distance run by the runner ? **[2]**

37. Read the case study given below and answer the questions that follow : **[4]**

A brooch is a decorative piece often worn on clothing like jackets, blouses or dresses to add elegance. Made from precious metals and decorated with gemstones, brooches come in many shapes and designs.
One such brooch is made with silver wire in the form of a circle with diameter 35 mm. The wire is also used in making 5 diameters which divide the circle into 10 equal sectors as shown in the figure.



(37.1) Find the central angle of each sector. **[1]**

(37.2) Find the length of the arc ACB. **[1]**

(37.3) (a) Find the area of each sector of the brooch. **[2]**

OR

(37.3) (b) Find the total length of the silver wire used. **[2]**

38. Read the case study given below and answer the questions that follow : **[4]**

BINGO is game of chance. The host has 75 balls numbered 1 through 75. Each player has a BINGO card with some numbers written on it. The participant cancels the number on the card when called out a number written on the ball selected at random. Whosoever cancels all the numbers on his/her card, says BINGO and wins the game.
The table given below, shows the data of one such game where 48 balls were used before Tara said 'BINGO'.

Numbers announced	Number of times
0-15	8
15-30	9
30-45	10
45-60	12
60-75	9

(38.1) Write the median class. **[1]**

(38.2) When first ball was picked up, what was the probability of calling out an even number ? **[1]**

(38.3) (a) Find median of the given data. **[2]**

OR

(38.3) (b) Find mode of the given data. **[2]**

- End of Question Paper -

Sample Paper 5

CLASS 10 | MATHEMATICS (STANDARD)

Board Examination Practice Paper (2026-27 pattern) | Level: Hard

Maximum Marks: 80

Time Allowed: 3 Hours

GENERAL INSTRUCTIONS

1. This question paper contains **38** questions. **All questions are compulsory.**
2. This question paper is divided into **five** sections - Section A, B, C, D and E.
3. **Section A:** Q. Nos. 1 to 18 are Multiple Choice Questions (MCQs) and Q. Nos. 19 and 20 are Assertion-Reason based questions of **1 mark** each.
4. **Section B:** Q. Nos. 21 to 25 are Very Short Answer (VSA) type questions carrying **2 marks** each.
5. **Section C:** Q. Nos. 26 to 31 are Short Answer (SA) type questions carrying **3 marks** each.
6. **Section D:** Q. Nos. 32 to 35 are Long Answer (LA) type questions carrying **5 marks** each.
7. **Section E:** Q. Nos. 36 to 38 are case study based questions carrying **4 marks** each. Internal choice is provided in one sub-part of the case study questions.
8. There is no overall choice. However, an internal choice has been provided in 2 questions in Section B (Q21, Q25), 2 in Section C (Q29, Q30), 2 in Section D (Q33, Q34) and 3 in Section E (Q36, Q37, Q38).
9. Draw neat diagrams wherever required. Take **$\pi = 22/7$** wherever required, if not stated.
10. Use of calculator is **NOT** allowed. Marks for each question are shown in square brackets on the right.

SECTION A - Multiple Choice and Assertion-Reason Questions (20 Marks)

1. There are two sections A and B of Grade X. There are 28 students in Section A and 30 students in Section B. What is the minimum number of books you will acquire for the class library so that they can be distributed equally among students of Section A or Section B ? **[1]**

(A) 144 **(B)** 2

(C) 420 **(D)** 272

2. Which of the following is a quadratic polynomial having zeroes $\frac{-2}{3}$ and $\frac{2}{3}$? **[1]**

(A) $4x^2 - 9$ **(B)** $\frac{4}{9}(9x^2 + 4)$

(C) $x^2 + \frac{9}{4}$ **(D)** $5(9x^2 - 4)$

3. If α and β are the zeroes of polynomial $3x^2 + 6x + k$ such that $\alpha + \beta + \alpha\beta = -\frac{2}{3}$, then the value of k is : **[1]**

(A) - 8 **(B)** 8

(C) - 4 **(D)** 4

4. If x = 1 and y = 2 is a solution of the pair of linear equations 2x - 3y + a = 0 and 2x + 3y - b = 0, then : **[1]**

(A) a = 2b **(B)** 2a = b

(C) a + 2b = 0 **(D)** 2a + b = 0

5. The pair of linear equations $(3x)/2 + (5y)/3 = 7$ and $9x + 10y = 14$, is : **[1]**

(A) consistent **(B)** inconsistent

(C) consistent with one solution **(D)** consistent with many solutions

6. The least positive value of k, for which the quadratic equation $2x^2 + kx - 4 = 0$ has rational roots, is **[1]**

(A) $\pm 2\sqrt{2}$ **(B)** 2

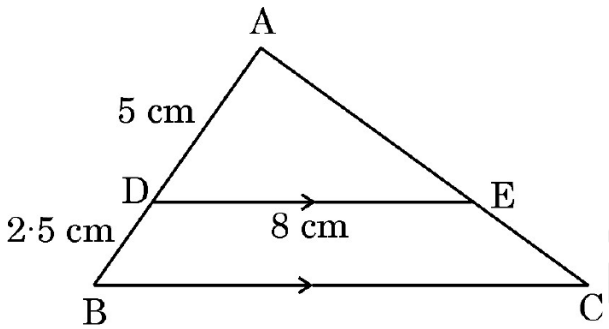
(C) ± 2 **(D)** $\sqrt{2}$

7. Which of the following sequence is *not* an A.P. ? **[1]**

(A) $2, \frac{5}{2}, 3, \frac{7}{2}, \ldots$ **(B)** - 1.2, -3.2, -5.2, -7.2, ...

(C) $\sqrt{2}, \sqrt{8}, \sqrt{18}, \ldots$ **(D)** $1^2, 3^2, 5^2, 7^2, \ldots$

8. In the given figure ΔABC is shown, in which DE ∥ BC. If AD = 5 cm, DB = 2.5 cm and DE = 8 cm, then the length of BC is : **[1]**



(A) 10 cm **(B)** 6 cm

(C) 12 cm **(D)** 7.5 cm

9. In what ratio, does x-axis divide the line segment joining the points A(3, 6) and B(-12, -3) ? **[1]**

(A) 1 : 2 **(B)** 1 : 4

(C) 4 : 1 **(D)** 2 : 1

10. AD is a median of ΔABC with vertices A(5, -6), B(6, 4) and C(0, 0). Length AD is equal to : **[1]**

(A) $\sqrt{68}$ units **(B)** $2\sqrt{15}$ units

(C) $\sqrt{101}$ units **(D)** 10 units

11. If 2 tan A = 3, then the value of $\frac{4 \sin A + 3 \cos A}{4 \sin A - 3 \cos A}$ is **[1]**

(A) $7/\sqrt{13}$ **(B)** $1/\sqrt{13}$

(C) 3 **(D)** does not exist

12. If sec θ - tan θ = m, then the value of sec θ + tan θ is : **[1]**

(A) $1 - \frac{1}{m}$ **(B)** $m^2 - 1$

(C) $\frac{1}{m}$ **(D)** - m

13. If $\alpha + \beta = 90°$ and $\alpha = 2\beta$, then $\cos^2 \alpha + \sin^2 \beta$ is equal to : **[1]**

(A) 0 **(B)** $\frac{1}{2}$

(C) 1 **(D)** 2

14. A piece of wire 20 cm long is bent into the form of an arc of a circle of radius $\frac{60}{\pi}$ cm. The angle subtended by the arc at the centre of the circle is : **[1]**

(A) 30° **(B)** 60°

(C) 90° **(D)** 50°

15. Area of a segment of a circle of radius 'r' and central angle 60° is : **[1]**

(A) $(\pi r^2)/2 - \frac{1}{2}r^2$ **(B)** $(2\pi r)/4 - (\sqrt{3})/4\ r^2$

(C) $(\pi r^2)/6 - (\sqrt{3})/4\ r^2$ **(D)** $(2\pi r)/4 - r^2 \sin 60°$

16. A hemispherical bowl is made of steel of thickness 1 cm. The outer radius of the bowl is 6 cm. The volume of steel used (in cm^3) is : **[1]**

(A) $182\ \pi$ **(B)** $\frac{182}{3}\ \pi$

(C) $\frac{682}{3}\ \pi$ **(D)** $\frac{364}{3}\ \pi$

17. The distribution below gives the marks obtained by 80 students on a test : **[1]**

Marks	Less than 10	Less than 20	Less than 30	Less than 40	Less than 50	Less than 60
Number of Students	3	12	27	57	75	80

The modal class of this distribution is :

(A) 10 – 20 **(B)** 20 – 30

(C) 30 – 40 **(D)** 50 – 60

18. From the data 1, 4, 7, 9, 16, 21, 25, if all the even numbers are removed, then the probability of getting at random a prime number from the remaining is : **[1]**

(A) $\frac{2}{5}$ **(B)** $\frac{1}{5}$

(C) $\frac{1}{7}$ **(D)** $\frac{2}{7}$

Assertion-Reason based questions. *Directions* : In Q. Nos. 19 and 20, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.

19. **Assertion (A) :** If the graph of a polynomial touches x-axis at only one point, then the polynomial cannot be a quadratic polynomial. **[1]**
Reason (R) : A polynomial of degree n(n >1) can have at most n zeroes.

20. **Assertion (A) :** The surface area of the cuboid formed by joining two cubes of sides 4 cm each, end-to-end, is 160 cm^2. **[1]**
Reason (R) : The surface area of a cuboid of dimensions $l \times b \times h$ is $(lb + bh + hl)$.

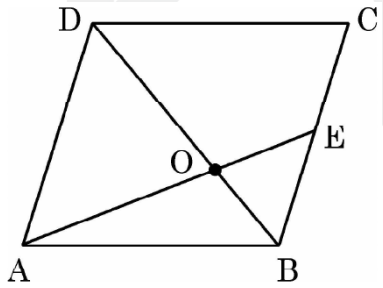
SECTION B - Very Short Answer Questions (10 Marks)

21. (a) Prove that $5 - 2\sqrt{3}$ is an irrational number. It is given that $\sqrt{3}$ is an irrational number. **[2]**

OR

(b) Show that the number $5\times11\times17+3\times11$ is a composite number.

22. In the given figure, ABCD is a parallelogram. AE divides the line segment BD in the ratio 1 : 2. If BE = 1.5 cm, then find the length of BC. **[2]**



23. The coordinates of the centre of a circle are (2a, a - 7). Find the value(s) of 'a' if the circle passes through the point (11, -9) and has diameter $10\sqrt{2}$ units. **[2]**

24. If the points A(4, 5), B(m, 6), C(4, 3) and D(1, n) taken in this order are the vertices of a parallelogram ABCD, then find the values of m and n. **[2]**

25. (a) If $\sin\theta + \sin^2\theta = 1$, then prove that $\cos^2\theta + \cos^4\theta = 1$. **[2]**

OR

(b) If $\tan\theta = 1/\sqrt{7}$, then show that $\frac{\text{cosec}^2\theta - \sec^2\theta}{\text{cosec}^2\theta + \sec^2\theta} = \frac{3}{4}$.

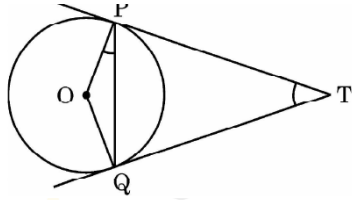
SECTION C - Short Answer Questions (18 Marks)

26. Three sets of Physics, Chemistry and Mathematics books have to be stacked in such a way that all the books are stored subject-wise and the height of each stack is the same. The number of Physics books is 144, the number of Chemistry books is 180 and the number of Mathematics books is 192. Assuming that the books are of same thickness, determine the number of stacks of Physics, Chemistry and Mathematics books. **[3]**

27. Jaya scored 40 marks in a test getting 3 marks for each correct answer and losing 1 mark for each incorrect answer. Had 4 marks being awarded for each correct answer and 2 marks were deducted for each incorrect answer then Jaya again would have scored 40 marks. How many questions were there in the Test ? **[3]**

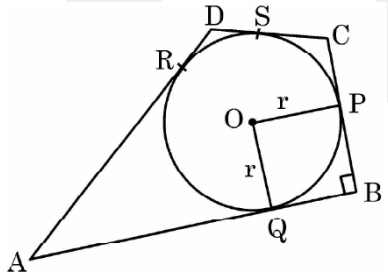
28. Prove that : $\frac{\tan\theta + \sec\theta - 1}{\tan\theta - \sec\theta + 1} = (1 + \sin\theta)/\cos\theta$ **[3]**

29. (a) Two tangents TP and TQ are drawn to a circle with centre O from an external point T. Prove that ∠PTQ = 2∠OPQ. **[3]**



OR

(b) In the given figure, a circle is inscribed in a quadrilateral ABCD in which ∠B = 90°. If AD = 17 cm, AB = 20 cm and DS = 3 cm, then find the radius of the circle.

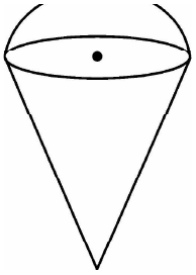


Career Ka Dost
Ask before you decide

30. (a) A room is in the form of cylinder surmounted by a hemi-spherical dome. The base radius of hemisphere is one-half the height of cylindrical part. Find total height of the room if it contains $(\frac{1408}{21})$ m^3 of air. (Take $\pi = 22/7$) **[3]**

OR

(b) An empty cone is of radius 3 cm and height 12 cm. Ice-cream is filled in it so that lower part of the cone which is $(\frac{1}{6})^{th}$ of the volume of the cone is unfilled but hemisphere is formed on the top. Find volume of the ice-cream. (Take $\pi = 3.14$)



31. The difference between the outer and inner radii of a hollow right circular cylinder of length 14 cm is 1 cm. If the volume of the metal used in making the cylinder is 176 cm^3, find the outer and inner radii of the cylinder. **[3]**

SECTION D - Long Answer Questions **(20 Marks)**

32. Vijay invested certain amounts of money in two schemes A and B, which offer interest at the rate of 8% per annum and 9% per annum, respectively. He received ₹ 1,860 as the total annual interest. However, had he interchanged the amounts of investments in the two schemes, he would have received ₹ 20 more as annual interest. How much money did he invest in each scheme ? **[5]**

33. (a) In a flight of 600 km, an aircraft slowed down its speed due to bad weather. Its average speed for the trip reduced by 200 km/h from its usual speed and time of flight increased by 30 minutes. Find the scheduled duration of the flight. **[5]**

OR

(b) Two pipes are used to fill a swimming pool. If the pipe of the larger diameter is used for 4 hours and the pipe of the smaller diameter for 9 hours, only half of the pool can be filled. Find how long it would take for each pipe to fill the pool, separately, if the pipe of smaller diameter takes 10 hours more than the pipe of larger diameter to fill the pool.

34. (a) Sides AB and AC and median AM of a ΔABC are proportional to sides DE and DF and Median DN of another ΔDEF. Show that ΔABC ~ ΔDEF. **[5]**

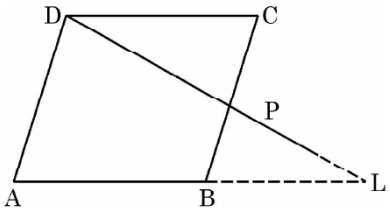
OR

(b) ABCD is a parallelogram, P is a point on side BC and DP when produced meets AB produced at L. Prove that

(i) $\frac{DP}{PL} = \frac{DC}{BL}$

(ii) $\frac{DL}{DP} = \frac{AL}{DC}$

(iii) If LP : PD = 2 : 3 then find BP : BC.



35. An SBI health insurance agent found the following data for distribution of ages of 100 policy holders. The health insurance policies are given to persons of age 15 years and onwards, but less than 60 years. **[5]**

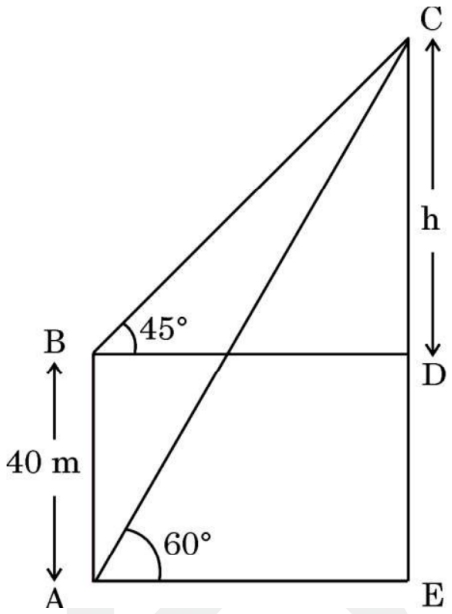
Age (in yrs)	Number of policy holders
15 – 20	2
20 – 25	4
25 – 30	18
30 – 35	21
35 – 40	33
40 – 45	11
45 – 50	3
50 – 55	6
55 – 60	2

Find the modal age and median age of the policy holders.

SECTION E - Case Study Based Questions (12 Marks)

36. Read the case study given below and answer the questions that follow : **[4]**

Amrita stood near the base of a lighthouse, gazing up at its towering height. She measured the angle of elevation to the top and found it to be 60°. Then, she climbed a nearby observation deck, 40 metres higher than her original position and noticed the angle of elevation to the top of lighthouse to be 45°.



(36.1) If CD is h metres, find the distance BD in terms of 'h'. **[1]**

(36.2) Find distance BC in terms of 'h'. **[1]**

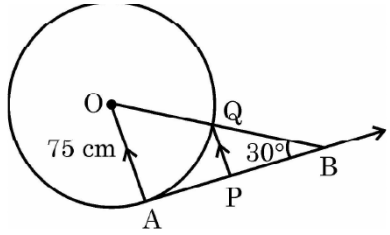
(36.3) (a) Find the height CE of the lighthouse [Use $\sqrt{3} = 1.73$] **[2]**

OR

(36.3) (b) Find distance AE, if AC = 100 m. **[2]**

37. Read the case study given below and answer the questions that follow : **[4]**

The discus throw is an event in which an athlete attempts to throw a discus. The athlete spins anti-clockwise around one and a half times through a circle, then releases the throw. When released, the discus travels along tangent to the circular spin orbit.
In the given figure, AB is one such tangent to a circle of radius 75 cm. Point O is centre of the circle and ∠ABO = 30°. PQ is parallel to OA.
Based on above information :



(37.1) find the length of AB. **[1]**

(37.2) find the length of OB. **[1]**

(37.3) (a) find the length of AP. **[2]**

OR

(37.3) (b) find the length of PQ. **[2]**

38. Read the case study given below and answer the questions that follow : **[4]**

India meteorological department observes seasonal and annual rainfall every year in different sub-divisions of our country.
It helps them to compare and analyse the results. The table given below shows sub-division wise seasonal (monsoon) rainfall (mm) in 2018 :

Rainfall (mm)	Number of Sub-divisions
200-400	2
400-600	4
600-800	7
800-1000	4
1000-1200	2
1200-1400	3
1400-1600	1
1600-1800	1

(38.1) Write the modal class. **[1]**

(38.2) (a) Find the median of the given data. **[2]**

OR

(38.2) (b) Find the mean rainfall in this season. **[2]**

(38.3) If sub-division having at least 1000 mm rainfall during monsoon season, is considered good rainfall sub-division, then how many sub-divisions had good rainfall ? **[1]**

- End of Question Paper -

Career Ka Dost
Ask before you decide

All the Best!

You have done the hard work. Now go and show it on the paper.

Exam-day reminders

1	Read the whole paper once and start with the section you are most confident about.
2	Give about 40 minutes to Section A, 20 to Section B, 35 to Section C, 50 to Section D, 25 to Section E, and keep 10 minutes for revision.
3	Write every step. In maths, steps carry marks even if the final answer goes wrong.
4	Draw neat figures in pencil wherever the question needs one, and label them fully.
5	Calculators are not allowed, so practise squares, cubes and square roots till they are quick.
6	Attempt every question. There is internal choice, so pick the one you know best.
7	Solve the 5 papers one by one, from Paper 1 (easy) to Paper 5 (hard), in exam conditions and revise from NCERT after each.

Career Ka Dost
Ask before you decide